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Key Genes, Altered Pathways and Potential Treatments for Muscle Loss in Astronauts and Sarcopenic Patients

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Abstract

Sarcopenia is characterized by loss of muscle mass and strength in the elderly. Interestingly, astronauts suffer from a sarcopenic-like phenotype due to microgravity, thus effective countermeasures and preventive strategies are needed. Earth precision medicine combined with statistical, co-expression network and pathway analysis enables us to explore gene expression data from people with and without sarcopenia to obtain a list of 21 Key Genes (KGs). We then validated our KGs upon data from human endothelial cells cultured in the International Space Station, and astronauts' samples from Japan Aerospace Exploration Agency and Inspiration 4 mission. Our results suggest that POMC and GOLGA8R are the most robust biomarkers identified for muscle loss. Finally, a pharmacological screening performed to target our KGs showed that POMC activity can be modulated using phase IV or approved drugs. Combining Earth's precision medicine with space data is a promising approach to address common conditions related to accelerated aging.

1. Introduction

Humanity has the objective to go to space and explore new worlds. During the last 60 years we have improved our technology to go beyond our planet's atmosphere, being able to transport human crew into orbit, set foot on the moon, and establish the International Space Station (ISS). These immense achievements served to gain experience and increase our knowledge about space hazards and how to countermeasure them. Despite our advancements in space technology, maintaining the health of astronauts during long-term missions remains a challenge. The permanence of astronauts in microgravity has gone from 100 min for Yuri Gagarin to more than 1 year inside the ISS [1]. Having longer space missions implies an increased risk of exposure to microgravity and radiation, and these factors lead to abnormal changes in cell behavior, a decrease in muscle mass and loss of bone density. These common health issues for astronauts affect their body's capacity to heal and adapt during a spaceflight and after returning to earth [2,3]. Therefore, much more research still needs to be done to generate preventive strategies to treat our bodies before, during and after long-term missions.

Sarcopenia is a disorder of the musculoskeletal system characterized by a decline in muscle mass and muscle quality that correlates with adverse outcomes such as falls, physical disability and mortality. Its etiology is complex and not fully understood. The condition is defined as "primary sarcopenia" when there is no other identifiable cause besides the aging process, while the term "secondary sarcopenia" is used when other causal factors (e.g., disease, malnutrition, sedentary condition) are clearly identified. Inflammatory and neurological disorders have a key role in the development of this condition, underlining a possible development of the disease in younger patients [4]. This condition is closely related to another nosological entity defined as frailty, a clinical syndrome which is characterized by a reduced resilience to stressors such as acute diseases and surgery, due to a progressive impairment in the function of different physiological systems. Aging is considered the biological background for the development of this condition. Sarcopenia's physical parameters of reduced physical performance, muscle strength and mass are central in the frailty syndrome; however, until now sarcopenia and frailty are considered as two

separate things, where the first one is mostly concerned with the loss of muscle tissue, while the second one is considered as a broad clinical syndrome that involves multiple organs, as well as cognitive and social elements [4,5].

The exposure to the space environment has a significant impact on the musculoskeletal system and causes a sarcopenia-like phenotype potentially contributing to the development of frailty-like condition in astronauts. Muscle loss causes cardiovascular deconditioning, and immune deficiencies, among other conditions in astronauts and in the elderly on Earth [2,6]. Several countermeasures to muscle loss in space have been studied and showed promising results such as constant exercise, extravehicular activities and adjustments in food intake to match astronauts requirements [1,7]. Having a better understanding of what drives muscle loss, especially in terms of gene expression patterns and deregulated cellular pathways, is critical in order to develop new and more effective therapies to prevent excessive muscle loss or even stop it.

Two factors have been evidenced to affect humans during aging and during spaceflight. First, in older humans there's a decrease in appetite and the estimated calorie intake is reduced by 1000 to 1200 kcal in men and by 600 to 800 kcal in women, using cohort and cross-sectional surveys [8–10]. Decrease in appetite has been seen from Apollo through the Space Shuttle Program, dietary intakes during flight averaged about 70% of predicted requirements, and intakes on the ISS have averaged about 80% of requirements [11–13] (Smith et al., 2021). Second, there is a decrease in insulin sensitivity during aging [14,15] as well as spaceflight and bedrest [13,16].

It has been observed that targeting myostatin, activin A and their associated pathways in mice at the ISS, protects them against muscle and bone loss in microgravity [3]. Interestingly, similar therapeutic approaches as those tested in mice and humans in space, are being used and further developed to provide methods and pharmacological interventions to overcome sarcopenia. Performing cellular, molecular and informatic analysis of sarcopenic patients could help to identify key genes and expression patterns significantly correlated with the disease. These genes and patterns may be useful biomarkers for muscle loss. Sarcopenia biomarkers applied to astronauts may help to prevent muscle loss, by implementing personalized physical, nutritional and even genetic support. As mentioned before, targeting myostatin expression or using antibodies to disable proteins related with the progression of the sarcopenia may prevent muscle loss during space travel.

The use of bioinformatics to identify potential pharmacological compounds to tackle aging-related diseases is a plausible option to generate preclinical and clinical evidence [17,18]. However, to the best of our knowledge, such an approach has not been applied to finding existing drugs as potential candidates to treat muscle loss, in both, older adults with sarcopenia and astronauts. Therefore, we apply here a bioinformatic analysis to identify genes that evidenced significant differences in gene expression data coming from people with and without sarcopenia. We then perform a gene co-expression network analysis, GO term enrichment analysis and KEGG pathway analysis to discover key genes (KGs) and pathways closely related to sarcopenia. Subsequently, we seek to validate the KGs and pathways

identified on data coming from: *i.* human umbilical vein endothelial cells (HUVEC) cultured on the ISS, *ii.* plasma cell-free RNA derived from blood samples of six astronauts collected before, during and after their visit to the ISS and *iii.* the Inspiration 4 Mission (I4) immune cells.

This study intends to identify and validate genes that are altered in patients suffering from muscle loss, by analyzing gene expression data from multiple sources. We also suggest potential pharmacological candidates, whose ability to tackle muscle loss can be further analyzed clinically.

2. Results

After studying databases from 118 people with and without sarcopenia (GSE111006, GSE111010 and GSE111016) (Migliavacca 2019), we identified 6892 DEGs* through statistical inference and estimated their individual predictive power for that condition by means of a simple classifier, i.e., k-nearest neighbors (Pedregosa et al., 2011).

Then, via co-expression network analysis upon those DEGs*, we identified the module with the highest correlation with sarcopenia, which we denote as BROWN (Fig. 1).

In order to identify linear relationships between DEGs75, we proceed to visualize the resulting Pearson's correlation coefficients in the form of a heatmap shown in Fig. 2. In this respect, the DEGs75 list contains 62 protein-coding genes, 26 pseudogenes and 23 lncRNAs as the best predictors of sarcopenia. Thus, the ability to predict sarcopenia through the DEGs75 highlights the importance of these non-coding genes as biomarkers.

The eigengene from the BROWN module obtained a Pearson correlation coefficient of 0.93 with sarcopenia. Subsequently, we performed pathway and GO analysis upon the 21 genes resulting from the overlapping of BROWN, NETWORK, DEGs* and DEGs** and they all significantly enriched in BP related to sarcopenia, see Figs. 4,5,6. Here, we denote this set of genes as KGs.

The genes in BROWN were mostly downregulated. Only three genes, namely APELA (Apelin Receptor Early Endogenous Ligand), AC023347.2 and NOX1 (NADPH Oxidase 1), had a Fold Change (FC) greater than 1.5, whereas 83 genes had a FC < 0.75. The top downregulated genes were LOC102724594, MT-TQ (Mitochondrially Encoded TRNA-Gln), AC015967.2, AC005014.3, MT-TE (Mitochondrially Encoded TRNA-Glu) and MFSD6L (Major Facilitator Superfamily Domain Containing 6 Like) with a FC between 0.59 and 0.68.

Additionally, we tested the nutritional gene expression biomarkers currently used to diagnose sarcopenia and found their MAS [19]. Not all of the expected genes were differentially expressed, but we found that bone morphogenetic proteins (BMPs) had the highest MAS among the nutritional biomarkers (BMP3 72.28%, BMP6 71.63%, BMP6 71.64% and BMP7 69,41%). Some of the described negative regulators had lower scores such as TGFβ (TGFB1 61.36%), and activins A and B (67.09% and 65.19%) .

Using the 121 DEGS from NETWORK (NETWORK- DEGs*), we obtained enriched Biological Processes (BP) and Molecular Functions (MF) terms depicted in Fig. 4C and D respectively. Interestingly, when we studied our KGs more closely, we found that five of them (NOX1, MKRN3 (Makorin Ring Finger Protein 3), GATA5 (GATA Binding Protein 5), GLS2 (Glutaminase 2), and POMC (Proopiomelanocortin)) respond to cells' energy levels through TP53 (Tumor Protein P53) and indirectly through MTOR (Mechanistic Target of Rapamycin Kinase), SIRT1 (Sirtuin 1) and PRKAA1 (Protein Kinase AMP-Activated Catalytic Subunit Alpha 1) (Supplementary Fig. 5). In accordance with these results, the bioenergetic alterations could be the result of altered oxidative phosphorylation, mitochondrial energy homeostasis and mTOR signaling related to sarcopenia and muscle loss. On BP we found that the most significant pathways were protein polyubiquitination (GO:0000209) followed by parent terms such as protein ubiquitination (GO:0016567), and protein modification by small protein conjugation or removal (GO:0070647). The next highest enriched pathway was associated with the intrinsic apoptotic signaling pathway (GO:0097193).

Then, to elucidate the role of each KG we searched for their presence in the resulting enriched pathways. NOX1 was present in both of these main processes by being part of all the apoptotic-related pathways and part of a parent GO term to protein ubiquitination. Notably, the gene MKRN3 is directly involved in the ubiquitination of proteins and exclusively related to ubiquitination on our GO result (Fig. 4A). GLS2 and HNRNPCL1 (Heterogeneous Nuclear Ribonucleoprotein C Like 1) were also part of both main pathways, but were only present on the most ancient terms.

The same analysis was performed but now for MF and we found that the most enriched pathways (FDR < .05) were also closely related to protein ubiquitination. More specifically, the ubiquitin conjugating enzyme activity (GO:0061631) was the most enriched pathway. Like in BP, NOX1 and HNRNPCL1 were present in parent terms to the top enriched pathway, but MKRN3 was again present in the child terms (Fig. 4B).

Next, we compared the expression of KGs in both sarcopenic patients and samples from spaceflight. We found that SHD, (Src Homology 2 Domain Containing Transforming Protein D), SHCBP1L (SHC Binding And Spindle Associated 1 Like), MKRN3, GLS2, POMC, EFHB (EF-Hand Domain Family Member B), and STPG2 (Sperm Tail PG-Rich Repeat Containing 2) were downregulated in both spaceflight and sarcopenic conditions, however this change was only significant in sarcopenia (Fig. 4E).

We assessed the expression of our KGs using the OSD-52 dataset obtained from in vitro cultured HUVECs. However, the clustering analysis revealed that the GC replicate 3 had an expression pattern more similar to FLT 1 and 2, while FLT-3 was more similar to GC-1 and 2. Although none of the 21 genes were significantly downregulated, their expression levels tended to be lower (e.g., MKRN3, SHCBP1L, OR51M1 (Olfactory Receptor Family 51 Subfamily M Member 1), and POMC), as shown in Fig. 5. Furthermore, we conducted an analysis using cell-free epigenome RNA-seq data obtained from JAXA, which revealed that the expression of GATA5, HOXB13 (Homeobox B13), SHCBP1L, SHD, GLS2, and MKRN3 tended to increase after spaceflight. It is noteworthy that GATA5, which is known to be increased in patients with sarcopenia, was also found to be elevated in astronauts post-flight. These findings suggest that the in

vitro model was affected by spaceflight through different pathways, as the gene expression of the KGs did not appear to be altered. Nonetheless, our results indicate that a cell-free blood sample from astronauts could be used to identify deregulated genes that serve as biomarkers for muscle loss and sarcopenia.

After studying the broad perspective of the NETWORK-DEGs* (Fig. 4), we decided to analyze exclusively the KGs and compared these results to the previous GO enrichment. For BP (Fig. 6) the most relevant terms were related to oxidative stress, pH regulation, epithelial differentiation, integrin biosynthesis, and histamine regulation processes (p. value < 0.05 for BiNGO and p.value < 0.01 for GOstat). We note that there is an enrichment in the cardiac Endothelial to Mesenchymal Transition (EMT) as well as blood pressure regulation, both of which are pathways that involve POMC and NOX1. Additionally, the ROS metabolic process requires the interaction of NOX1, PRG3 and GLS2. These results point to the relevance of oxidative stress, cardiovascular alterations, and possibly a connection between both of these processes.

We performed the same analysis for MF and we found an interesting link between POMC and four MF enriched terms (Fig. 7). More specifically, the enriched terms were Superoxide generating NAD(P)H oxidase activity, and melanocortin receptor binding (type I and IV). In contrast to the results from GOstat, through BiNGO we observed that the regulation of hydrogen and proton channels seemed to be enriched. These results are relevant since the hydrogen ion channel activity, oxidoreductase and FAD activity may be related to the mitochondrial metabolism.

During spaceflight, astronauts are exposed to microgravity and other stressful factors that can affect the expression levels of KGs. To investigate these potential changes, we analyzed skin biopsies from the I4 data after spaceflight. We found that 16 out of the 21 KGs showed deregulation, and that the KGs were differentially expressed across the epidermal layers in the I4 data. HOXB13, NOX1, SHD, SHCBP1L, GJB4 (Gap Junction Protein Beta 4), and HNRNPCL1 were overexpressed in two of the four studied compartments (the inner epidermis and outer dermis). Furthermore, six KGs were overexpressed after the flight (OR51M1, STPG2, BFSP2 (Beaded Filament Structural Protein 2), and, to a lesser extent, HNRNPCL1, GJB4, and SHCBP1L) in the vascular compartment. Due to the changes observed in 16 of the KGs in the skin biopsies, this sample type could be informative about the deregulation of genes related to muscle loss in astronauts and the risk of sarcopenia (Fig. 8).

Blood cells from the I4 astronauts could reflect changes in the identified KGs and may represent a less invasive sample source to analyze genes associated with the risk of muscle loss and sarcopenia in astronauts and patients. Blood cells (B-cells, CD4 + and CD8+-Tcells, CD14 + Monocytes, Natural Killers, PBMCs, and other cells) showed different sets of deregulated KGs before and after spaceflight. Out of the 21 KGs, 12 showed significant levels of up and downregulation. The deregulated genes ordered from the most common to the less represented in the samples are: GLS2 (present in 7 blood cell types), STPG2 (6), EFHB (5), MKRN3 (5), BFSP2 (5), FSPI2 (5), GOLGA8AR (5), FAM71D (4), OR51M1 (3), NOX1 (3), SHD (3) and SHCBP1L (2). Interestingly, the majority of the analyzed KGs did not go back to preflight expression

levels. In addition, one day after returning to earth, most genes were upregulated representing a possible early stress response.

Based on the fold change values obtained for all KGs in the sarcopenia database, we determined the genes that suffered an increase (i.e., fold change > 1) or decrease (i.e., fold change < 1) in their expression values. We then used this information as a baseline to compare the expression changes upon the same KGs evidenced across data from I4 and JAXA. More specifically, the fold change of each KG in I4 data was computed using the average expression values collected 92 days before flight and 1 day after flight, whereas for JAXA, those expression values were collected 168 days before and 3 days after flight. To be consistent, in both cases we took the most distant measurements taken to the flight date and the closest after flight. Figure 10, shows in green the genes that changed their expression values according to the baseline, and thus in a certain way validate our findings. It is very interesting to see the robustness of POMC, as it behaved identically to our baseline, in terms of down and up regulation, in 9 out of 11 different datasets from JAXA and I4.

Finally, we investigated which KGs could be targeted and regulated by pharmacological agents in clinical trials or already in the market. To begin, we used our KGs to find chemical interactions from the Comparative Toxicogenomics Database (CTD) and found that 4 of them presented data linked to pharmacological interactions. These four genes are POMC, NOX1, GLS2 and SHD. The identified pharmacological agents have already reached at least Phase IV in clinical trials and could therefore be studied for drug repurposing and to possibly target the KGs.

Among the genes studied, POMC seems to be a robust biomarker of sarcopenia, according to Fig. 10, and thus blocking or suppressing its activity with Potassium, Sodium chloride, Lithium, Guanylyl Imidiophosphate and Guanosine Triphosphate could represent a good therapeutic option for people at risk.

3. Discussion

Space exploration has inspired human kind, prompting us to venture beyond the confines of Earth's boundaries and advance technological development. However, long-term missions pose a significant health risk to astronauts, necessitating careful consideration and understanding. The loss of muscle mass is a common effect of space conditions, especially microgravity, which acts as a driver of secondary sarcopenia-like phenotype [20]. Muscle loss impairs our capacity to regain physical activity, restore a healthy metabolic function and promotes the advancement of the hallmarks of aging [21]. By studying the deregulation of genes in sarcopenic patients on Earth, we can gain insights into the pathophysiology of muscle loss and use this information to develop strategies to prevent and mitigate the adverse effects experienced by astronauts.

We aimed to identify a list of biomarkers that can help diagnose early muscle loss in both people on Earth and astronauts, and propose potential pharmacological targets for further investigation. Our

findings suggest that the deregulation of the gene expression observed in sarcopenic patients on Earth, and subsequently validated through data from spaceflight missions samples, could imply bioenergetic alterations related to oxidative phosphorylation and mitochondrial energy homeostasis, leading to changes in mTOR signaling. These findings offer valuable insights that can guide the development of preventive or therapeutic strategies for muscle loss in both terrestrial and extraterrestrial environments.

Maintaining a balance between anabolism and catabolism is crucial to prevent muscle atrophy, which can lead to sarcopenia in humans. However, when comparing the basal rates of muscle protein synthesis and breakdown between young and old adults, the expected differences are not observed. This has led to the hypothesis that older adults are resistant to anabolic stimuli [22,23]. Furthermore, a recent study focusing on molecular signatures of men with sarcopenia compared to age-matched healthy individuals found that the main difference was in a low mitochondrial bioenergetic capacity, specifically oxidative phosphorylation and mitochondrial energy homeostasis. This study detected transcriptional signatures of an altered mTOR signaling, translation, and protein synthesis in human sarcopenia, which are consistent with the role of anabolic resistance in the decline of muscle mass and strength during aging [24].

Insulin and leptin are hormones that play a crucial role in energy balance signaling within cells. When there is energy surplus, insulin and leptin signal the cell by allowing FOXO1 phosphorylation which causes it to exit the nucleus and enables POMC transcription [25]. Conversely, when food intake is decreased and leptin and insulin sensitivity are reduced, FOXO1 remains in the nucleus, repressing POMC expression. Whereas lower levels of POMC elicit AMPK activation and positive regulation of cAMP signaling [26].

In particular, previous studies have shown that increased intake of leucine can lead to an increase in POMC mRNA expression in hypothalamic neurons [27–29]. While a previous microarray showed that POMC expression is specific to the pituitary [30], the results we analyzed from RNA sequencing show that POMC is being expressed in other tissues as well and tends to be downregulated in sarcopenia as well as after space flight. This data suggests that POMC could potentially serve as a biomarker for energy homeostasis in cells. In fact, previous research has proposed POMC as a biomarker for aging, as its expression was found to decrease in the hypothalamus of older rats after fasting compared to young rats [31].

POMC is a protein precursor that is cleaved to produce hormones such as adrenocorticotropic hormone (ACTH), β-lipotropin, and β-endorphin, among others, each of which exerts different functions in cells [32]. POMC expression has been reported in non-pituitary tissues in humans, including the skin and placenta. Interestingly, we found using the Human Protein Atlas that in skeletal muscle POMC has normalized transcripts per million values of 5.7, but is not detectable through immunohistochemistry [33]. The role of POMC overexpression in the hypothalamus has been related to muscle wasting; however, there is no data regarding POMC being secreted by myocytes [34,35]. In this work we observe that POMC expression seems

to be regulated by energy sensing pathways, showing to be downregulated in Sarcopenic samples from muscle biopsies and in immune cells from astronauts after space flight. As we define drugs that could regulate POMC expression, these should be further tested in order to reach the cells and tissues that may lead to muscle loss and a sarcopenic phenotype in people on Earth and astronauts.

When analyzing the enriched MF of KGs, we found a predicted and significant enrichment of hydrogen ion channel activity, oxidoreductase and FAD activity using BiNGO, and superoxide generating NAD(P)H oxidase activity using GOstat. These pathways are mainly related to the presence of NOX1 in our KGs. Interestingly, we again observed the relevance of POMC, this time through the more specific term of melanocortin receptor binding, which was found to be downregulated in patients with sarcopenia. Furthermore, as shown in Fig. 10, POMC seems to be the most robust biomarker, with downregulation observed in all except one of the I4 datasets, consistent with the changes seen in the sarcopenia database. Given that POMC can be regulated using available and relatively common pharmacological options, it represents an interesting therapeutic target for future research. Potential drugs that target POMCs may also have similar effects in the regulation of leptin and insulin sensitivity.

The results presented in Figs. 4, 5, and 6 show by gene ontology and network analysis that the deregulated KGs are involved in cell's catabolism, oxidative stress, and epithelial differentiation. This deregulation may lead to effects in the AMPK signaling pathway, which affects processes such as autophagy and proteasome activity. AMPK activation is a response to nutrient scarcity, which triggers homeostatic mechanisms such as the mobilization of internal stores through autophagy [26]. In addition, we evidence a lower expression of POMC in the samples shown in Fig. 10, including sarcopenic biopsies and immune cells of astronauts after spaceflight. These findings related to AMPK signaling and the suggested deregulations in anabolic resistance, the maintenance of homeostasis in musculoskeletal tissue [36–38]. Therefore, presenting autophagy while the tissue remains unable to meet a protein synthesis balance may explain muscle wasting. Overall, these results highlight the importance of further research into the regulation of AMPK signaling and POMC expression as potential therapeutic targets for preventing or treating muscle wasting.

Considering the bioenergetics deficiencies that arise during aging or as a result of spaceflight, there are several strategies that could be considered based on our results. For instance, Gómez et al (2021) mention the administration of antioxidants to prevent the damage attributed to space flight and that the dosage should consider the the genetic background of the person in Earth and space. However, the antioxidant combination and doses need to be further investigated [1]. In contrast, Lee et al. (2020) propose a different approach that focuses on inhibiting myostatin and activin A through ACVR2B/Fc decoy receptor [3]. Another potential strategy is the use of naloxone, which is one of the four approved drugs that can increase the activity of POMC.

HNRNPCL1 is a ribonucleoprotein (RNP) that may play a role in nucleosome assembly. Although HNRPLNC1 has not been directly linked to cellular energy states like POMC, there is evidence of intranuclear clustering/rearrangement of RNP structures in aging muscle samples compared to adult

samples [39]. Studies have also shown that myoblasts with impaired pre-mRNA maturation pathways differentiate into myotubes exhibiting structural defects similar to those of senescent healthy myotubes. These findings suggest that DM satellite cells have a reduced regeneration capability and may contribute to muscular dystrophy by generating defective myotubes [40].

Among the KGs analyzed, it has been observed that MKRN3 most directly interacts with protein ubiquitination [41]. However, MKRN3 does not have a described interaction with energy and the regulation of metabolism, except for AMPK ubiquitination [42]. GLS2, have a key role in the regulation of glutamine catabolism and promotes mitochondrial respiration. This leads to an increase in ATP generation by catalyzing the synthesis of glutamate and alpha-ketoglutarate. Interestingly, it also increases cellular antioxidant function via NADH and glutathione production [43]. Interestingly, it has been observed that P53 increases the GLS2 expression, augmenting glutamate and α-ketoglutarate presence, enhancing mitochondrial respiration and ATP generation. GLS2 increases glutathione levels and decreases the oxidative stress produced by ROS [43]. The GJB4 and GOLGA2 genes are structural proteins that play a key role in establishing cell-to-cell connections. For instance, GJB4 is part of the gap junctions; with related functions to promote propagation in the muscle tissue. Interestingly, impairing gap junctions results in reduced muscle force, more noticeably in the soleus, driven by activation of TGFβ/smad2/3 signaling and increased ROS levels [44]. While GOLGA2 links to microtubulin during mitosis, it is also required for protein transport. Mutations in GOLGA2 resulted in muscle fiber atrophy in humans [45,46].

Among the KGs, NOX1 was one of the most interesting findings in our study due to its high MAS and direct association with mitochondrial metabolism. Although NOX1 is not the primary form of NOX in skeletal muscle, it has been reported to play a crucial role in the proliferation of vascular smooth muscle. Moreover, NOX enzymes have been recognized as key regulators of skeletal muscle redox homeostasis under various health and disease conditions [47]. Overall, our results suggest that altered bioenergetics resulting from sustained protein degradation and autophagy may be the underlying cause of sarcopenia.

Altered bioenergetics may trigger the changes observed in the biological processes that were enriched using only the KGs. For instance, the reactive oxygen species metabolic process was associated with NOX1, PRG3, and GLS2, while the Superoxide metabolic process was associated with NOX1 and POMC (Fig. 6A-B). These processes are especially relevant considering the negative impact of ROS on cardiovascular function and its role in increasing the cardiovascular system's aging [1,48]. In blood vessels, nitric oxide (NO) is essential for reducing blood pressure by vasodilating vessels and increasing tissue oxygenation. However, the presence of ROS, peroxynitrite, and tetrahydrobiopterin can reduce physiological NO levels [49]. This may be related to the enrichment of histamine biosynthetic processes in Fig. 6C, and taken together with the enrichment of Cardiac Epithelial Mesenchymal Transition, the results seem to be associated with changes in the cardiovascular system. One recent study showed a shift in the response to NO levels as a mechanistic adaptation to space conditions observed in rats since vasodilation remained unchanged [50].

In conclusion, sarcopenia and microgravity-related muscle loss remain major health concerns, and effective countermeasures and preventive strategies are needed. Through the use of Earth's precision medicine and space data, we were able to identify a list of 21 Key Genes (KGs) associated with sarcopenia. Validation of our KGs in astronauts' samples from various missions confirmed the potential of POMC and GOLGA8R as robust biomarkers for muscle loss. Furthermore, pharmacological screening identified approved drugs that can modulate POMC activity. This innovative approach combining Earth's precision medicine with space data opens up new avenues for understanding and addressing common conditions related to accelerated aging.

4. Methodology

Protein-to-protein, as well as protein-to-DNA and RNA interactions, can be represented as networks, shedding light on the way our body maintains its health or develop diseases [51]. However, before building such networks, a list of genes is necessary and is usually obtained through the analysis of gene expression data, which has become widely accessible. Here, we make use of such data in our quest to find genes that may play an important role in the genesis of sarcopenia. We then validate the relevance of the genes identified against data obtained from Human Umbilical Vein Endothelial Cells (HUVEC), plasma cell-free RNA samples, blood cells and skin that were exposed to space conditions. Finally, we suggest potential pharmacological candidates that may regulate the altered genes to prevent or stop further muscle loss in the elderly and astronauts during space missions.

All experiments were conducted on an Intel(R) Xeon(R) Silver 4114 CPU 2.20 GHz with 80 GB of RAM running Ubuntu GNU/Linux, version 22.04.1 LTS. Each experiment may use up to 10 CPUs in parallel.

4.1 Genes and Pathways Related to Sarcopenia

Here, we used databases: (1) GSE111006 Hertfordshire Sarcopenia Study (HSS, 68–77 years old) (2) GSE111010 Jamaica Sarcopenia Study (JSS, 63–89 years old) and (3) GSE111016 Singapore Sarcopenia Study (SSS, 65–79 years old) that provide skeletal muscle transcriptome profilings derived from 32 and 86 sarcopenic and healthy men, respectively. The data was obtained by the authors [24] through expression profiling by high throughput sequencing (GSE111017) using the platform GPL16791 Illumina HiSeq 2500.

Prior to performing the following analysis, we applied the variance stabilizing transformation implemented on the package DESeq2 (Love et al, 2014) to the databases GSE111006, GSE111010 and GSE111016. In order to identify genes that evidence statistically significant differences (i.e., $p < .05$), we performed Mann–Whitney U tests (Mann and Whitney, 1947) on the corresponding samples of gene expression data for every single gene (i.e., 65217 genes) in a pair-wise manner in terms of their expression levels across samples from both sets of patients. The non-parametric test mentioned above was employed because the distributions of the samples analyzed do not fulfill the normality assumption. We denote the subset of genes resulting from this procedure as differentially expressed genes (DEGs*), i.e., genes with $p < .05$.

Related genes, in terms of regulation or interaction, often display similar expression profiles, and to find such gene modules, we used the package WGCNA [52] (Langfelder and Horvath, 2012) version 1.71 to develop a network analysis from expression data of 118 samples across all DEGs*. No excessive missing values nor outliers were identified on the data. By analyzing the network topology, we determined a soft thresholding power equal to 14, which we used to compute gene co-expression similarity and adjacency. That allows us to obtain the topological overlap matrix, which is necessary to get the different modules via hierarchical clustering using the dynamic tree-cut algorithm.

Moreover, we used a simple k-nearest neighbors classifier (Pedregosa et al., 2011) in order to estimate the individual predictive power of each gene in DEGs*. Given the small data set used here (i.e., 118 samples) it was not possible to build a classifier deploying sophisticated methods such as deep neural networks. The parameters of this classifier were tuned by using the package irace (López- Ibáñez et al, 2016) version 2.3. We measured the mean and the standard deviation of the accuracy score (i.e., percentage of correct predictions) across 100 independent experiments using a stratified 5-fold cross-validation on the data sets above mentioned. All DEGs* with a mean accuracy score higher or equal to 75% are denoted as DEGs75. The code used for the analysis mentioned so far can be found in 10.5281/zenodo.7411777.

Next, we used all genes present in the module with the highest correlation to sarcopenia, returned by the network analysis mentioned above, as input for Cytoscape to create a merged network, denoted as NETWORK, using public databases such as MINT [53], IntAct [54] and Uniprot [55] for human data exclusively. We compared the resulting groups and used Gene Ontology (GO) for genes in both DEGs** ($p < .01$) and DEGs* ($p < .05$) as well as genes present in NETWORK (Fig. 3). The resulting list of genes was used as input for GOSTAT (2022-11-03) to identify the BP and MF) associated with each of those genes for Fig. 7. Charts were plotted using the R package GOplot. Using a Venn diagram (from Venny 2.1 [56]) We determined the genes that were shared across the different gene lists and we denote them as KGs.

Additionally, we investigated the relevance of the KGs by themselves (using GOstat and BinGO from Cytoscape) and compared these results to the previous GO analysis. Again we used R to plot our results.

4.2 Validating biomarkers using samples that have been subjected to space conditions

4.2.1 Human Umbilical Vein Endothelial Cells

We assessed the expression of KGs on six samples (three for spaceflight and three for 1g) of Human Umbilical Vein Endothelial Cells (HUVEC), i.e., OSD-52 HUVECs, cultured for 7 days on the ISS. This information was obtained from Open Science Data Repository, formerly known as GeneLab Datasets. For this database, gene expression was analyzed by the authors using a cDNA microarray analysis using Affymetrix Gene Human 1.0 ST Arrays. The expression heatmap plots and dendrogram were obtained using the package “ComplexHeatmap” on R [57]

4.2.2 Inspiration 4 (I4) Mission

Methods for generating and processing the single-cell immune profiling are described in companion manuscripts. (Overbey et al., *Nature*. *Under review*. Kim et al., Nature. *Submitted*). Datasets have been uploaded to two data repositories: the NASA Open Science Data Repositories (OSDR; osdr.nasa.gov; comprised of GeneLab 67 and the Ames Life Sciences Data Archive [ALSDA] 68 2), and the TrialX database (OSDR-570). The KGs have been used for identifying DEGs from the I4 immune cells. Seurat was used to normalize RNA count data and calculate average expression of each gene. The Seurat FindMarker function was used to calculate I4 DEGs ($P < .05$). Average expression of those genes was used to generate heatmaps, which were created with the heatmap R package. For skin spatial transcriptomics, skin biopsies were obtained from all I4 crew members, once before flight and as soon as possible after return (L-44 and R + 1). These biopsies were flash frozen and processed with the NanoString GeoMx platform. A 20x scan was used to select 95 freeform regions of interest (ROIs) to guide selection of outer epidermal (OE), inner epidermal (IE), outer dermal (OD) and vascular (VA) regions. GeoMx WTA sequencing reads from NovaSeq6000 were compiled into FASTQ files corresponding to each ROI and converted to digital count conversion files using the NanoString GeoMx NGS DnD Pipeline. From the Q3 normalized count matrix that accounts for factors such as capture area, cellularity, and read quality, the DESeq2 method was used to perform differential expression analysis.

4.2.3 Plasma cell-free RNA samples from Japan Aerospace Exploration Agency

Plasma cell-free RNA samples were derived from blood samples from six astronauts collected before, during, and after the spaceflight on the ISS. These samples were later used to perform RNA-seq (Illumina NextSeq500) and their aggregated RNA differential expression data were shared through NASA's Open Science Data Repository (Accession Number OSD-530) [58]. Mean expression values were obtained from Standard Error of Means (SEM) normalization of read counts of all the astronauts for each time point.

4.3 Potential Pharmacological Candidates

We adopted a drug repurposing approach and investigated which of our KGs may be pharmacologically regulated. We used KGs as input for the Comparative Toxicogenomics Database (CTD) and obtained a dataset of chemical-gene interaction. We present only those chemicals whose results have been observed in humans. Finally, to determine the approved drugs we used ChEMBL and thus describe the clinical trial stage of each of the chemicals mentioned.

Declarations

Author contributions:

VC, JED, JP, AHV, JP, JKK, IS and AC wrote the manuscript and performed data analysis; VC, JED, AHV, JP, JKK, EO, IS, GN, ACAM, CHM, AB and AC data and manuscript revision; comments and editions, with significant input to its final form and approval; Conception AC, Conceptualization VC, JED and AC.

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Declaration of Competing Interest:

AC is the leader of the Dragon BioMed-USFQ initiative, and AC and VC are its founders. All the other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Figures

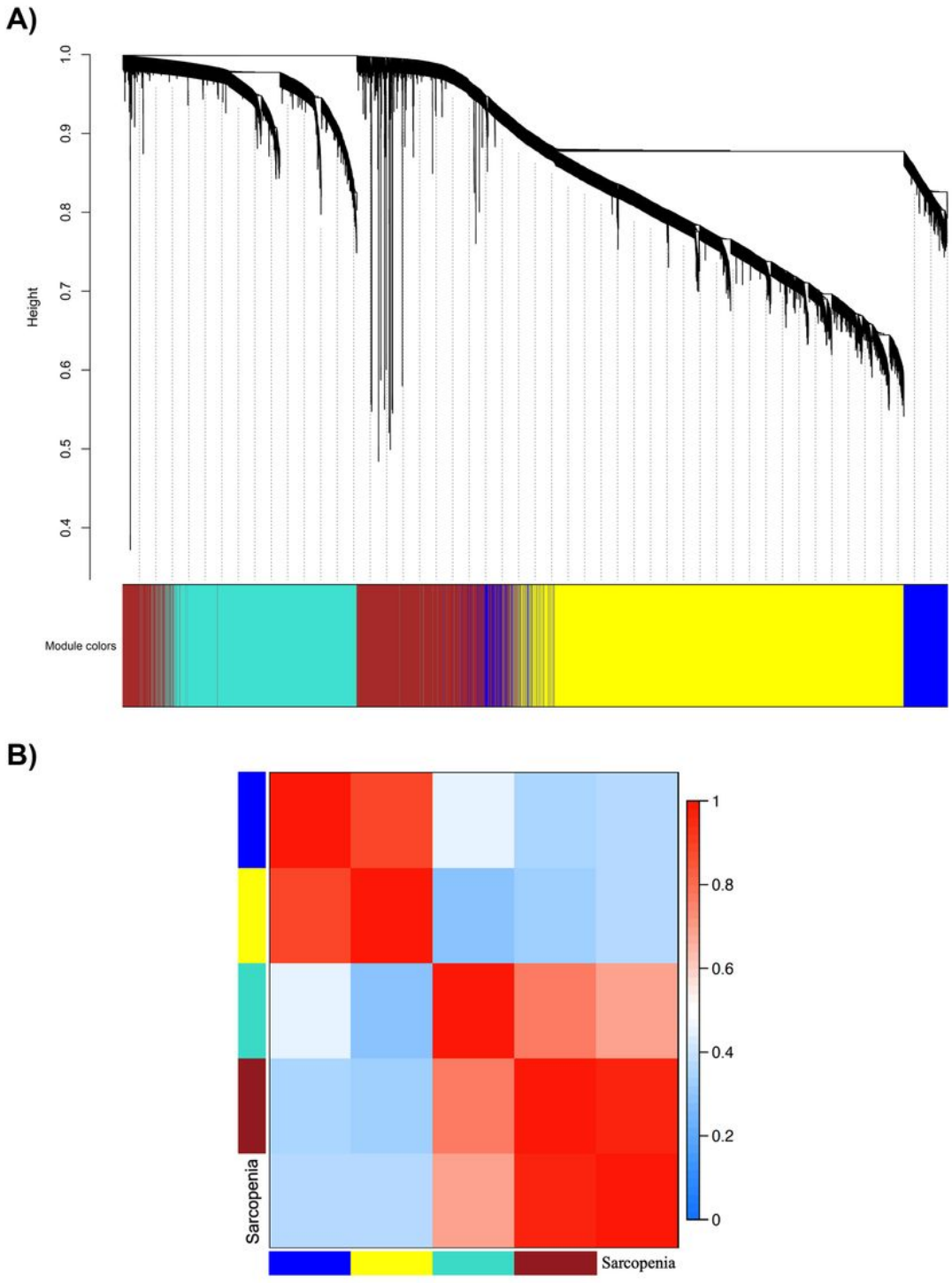


Figure 1

Weighted correlation network analysis of gene expression data of DEGs

A Cluster dendrogram of the different modules. **B** Pearson's correlation coefficients of Eigengenes and trait (i.e., sarcopenia) are shown in the form of a heatmap.

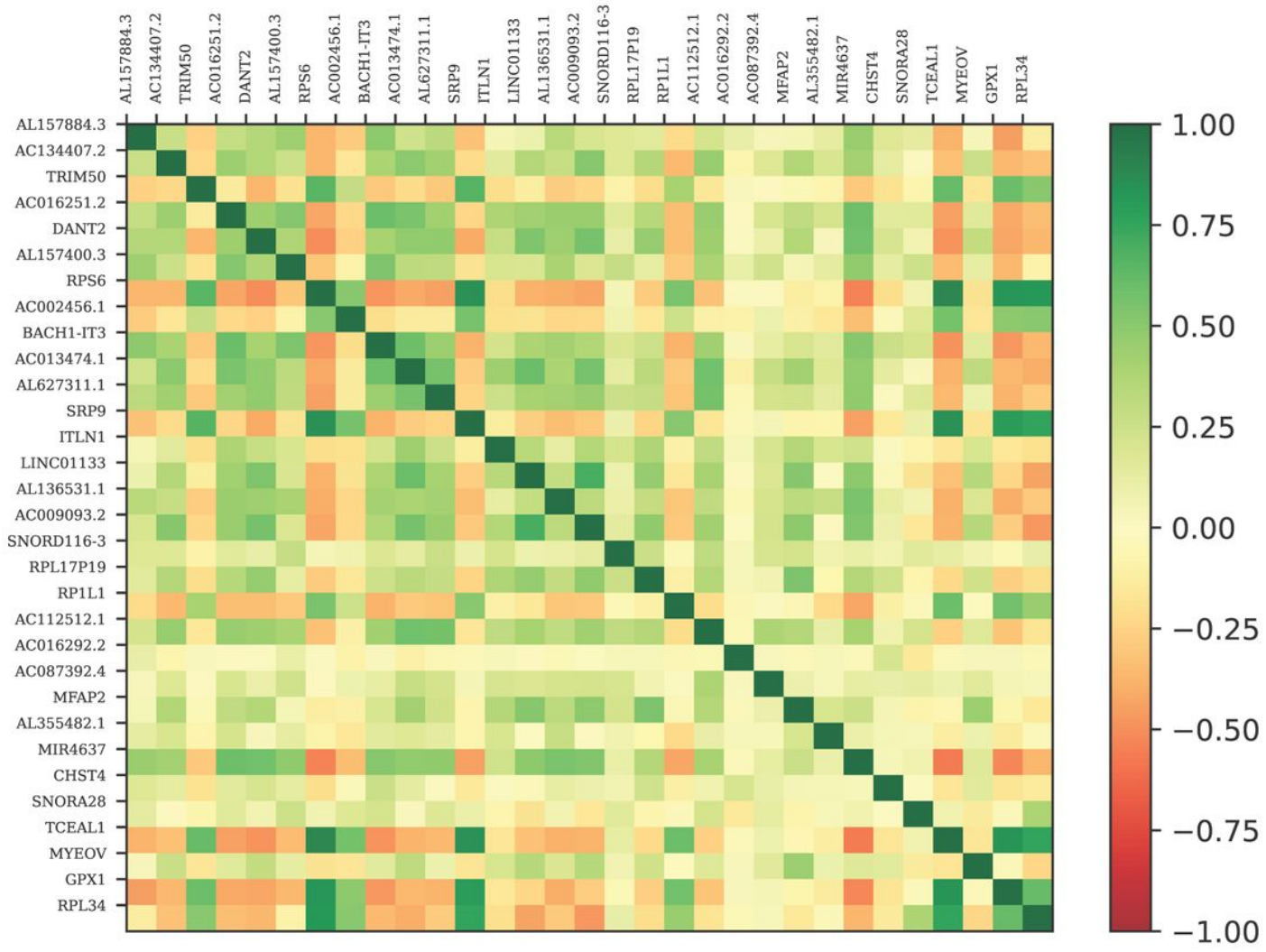


Figure 2

Heatmap of Pearson's correlation coefficients obtained from expression values of DEGs with an average accuracy score for sarcopenia $\geq 75\%$.

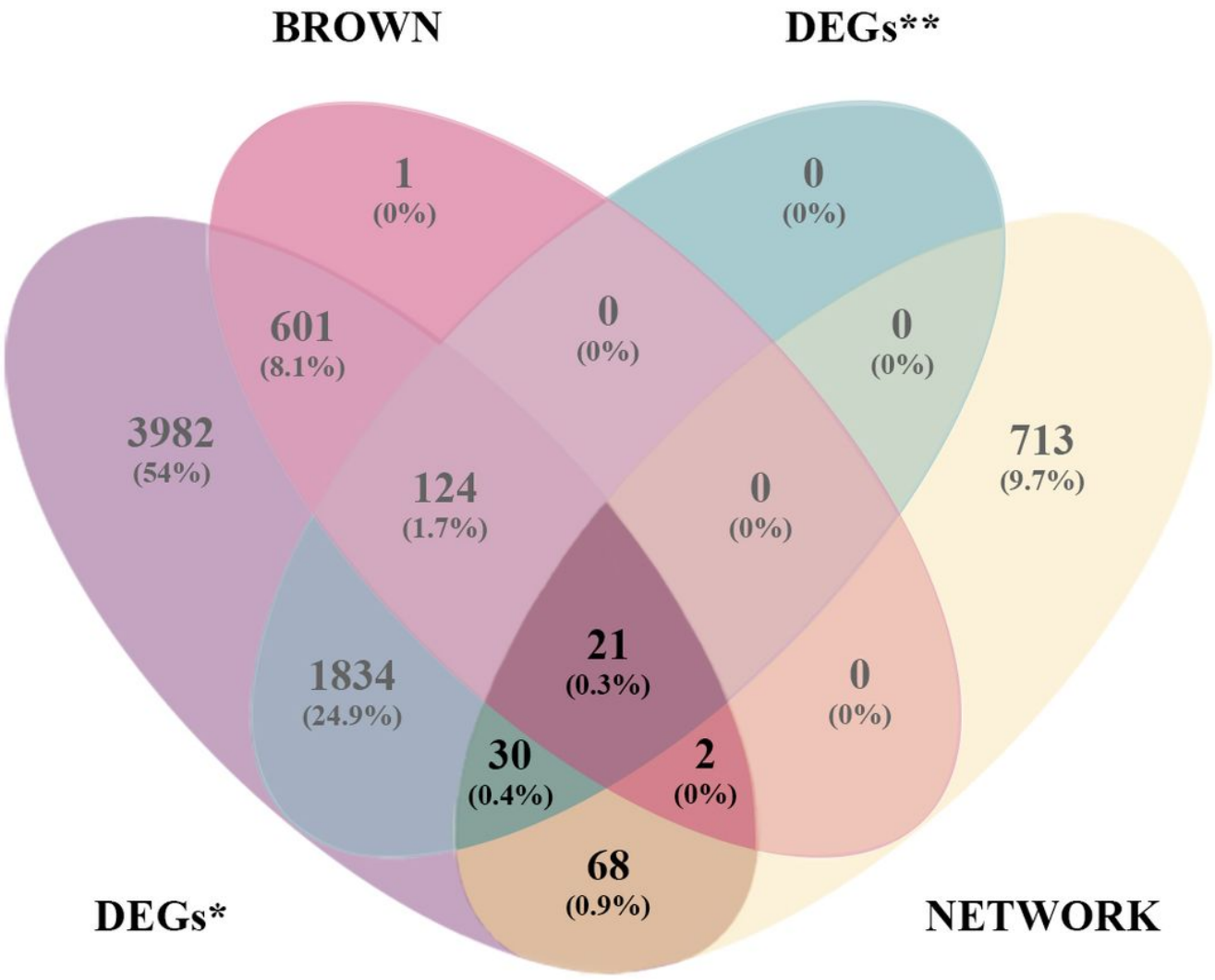


Figure 3

Venn diagram representing the different gene lists. The highlighted section (teardrop shape) has a total of 121 genes that were selected to the enrichment pathways with GO. BROWN, NETWORK, and DEGs** share 21 genes, which are denoted as KG in sarcopenia.

BROWN: Genes present in brown module obtained via network co-expression analysis

DEGs*: Differentially expressed genes, p<.05

DEGs:** Differentially expressed genes, p<.01

NETWORK: genes obtained via pathway analysis with Cytoscape using BROWN

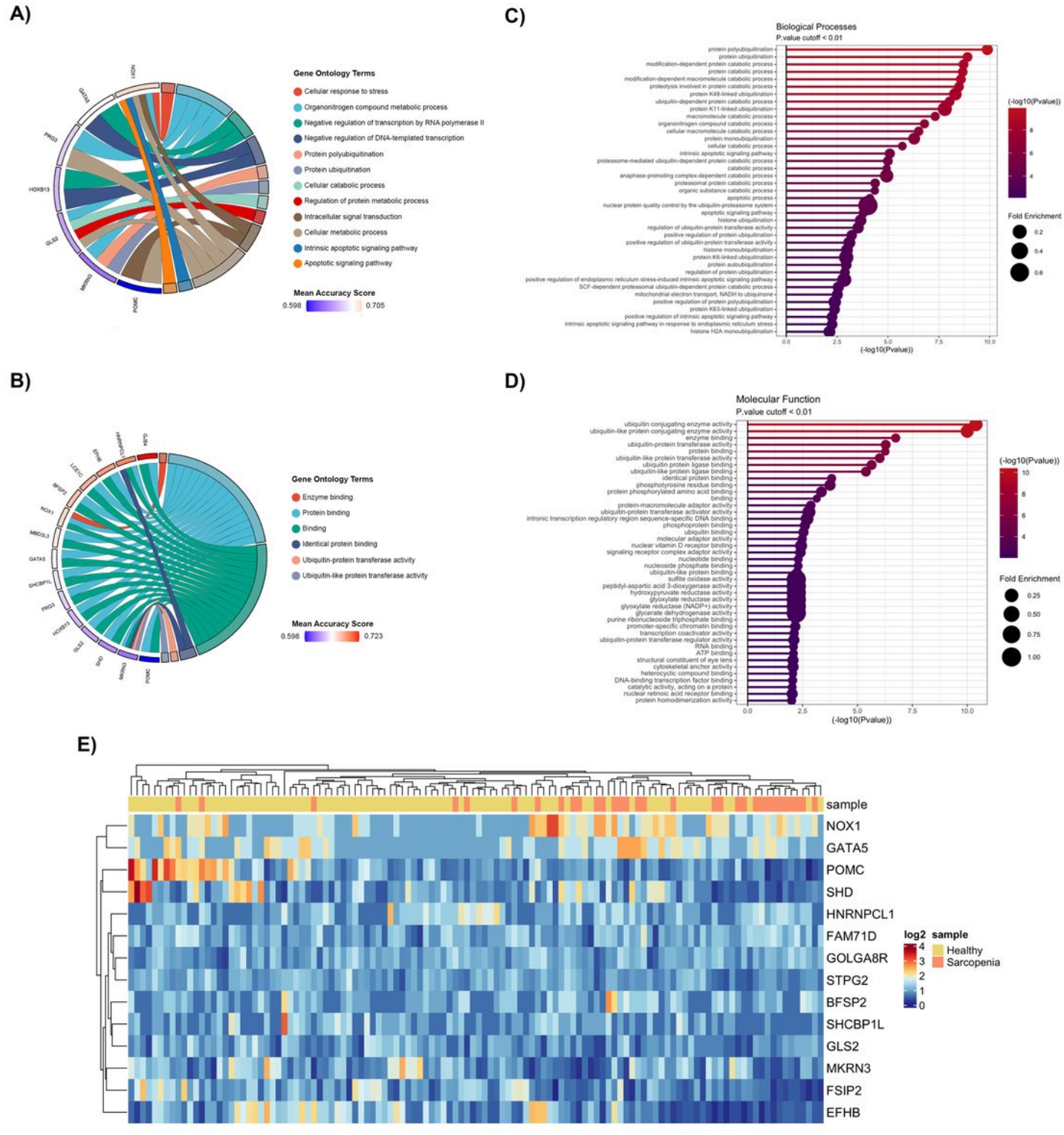


Figure 4

KG are relevant for catabolic processes using NETWORK-DEGs* as background

A The GOchord plot shows the relevance of our curated list of KGs in the significantly enriched BP GO terms. The colors on each gene represent their MAS, which is their ability to predict sarcopenia (See Supplementary Figure 1 for Hierarchy of the Gene Ontology Terms). **B** Shows the same information but

focusing on GO for MF (See Supplementary Figure 2 for Hierarchy of the Gene Ontology Terms). **C** Enriched Biological processes associated with either catabolism, ubiquitination or apoptosis, full list contains 305 terms. **D** Enriched Molecular Functions, full results shown. **E** Gene expression of KGs in the pooled subjects from GSE111006, GSE111010 and GSE111016. Out of the KGs, using 14 of them showed the best clustering to differentiate sarcopenic from healthy patients. The red and blue colors represent genes upregulated and downregulated respectively. The clustering is generated based on the euclidean distance method.

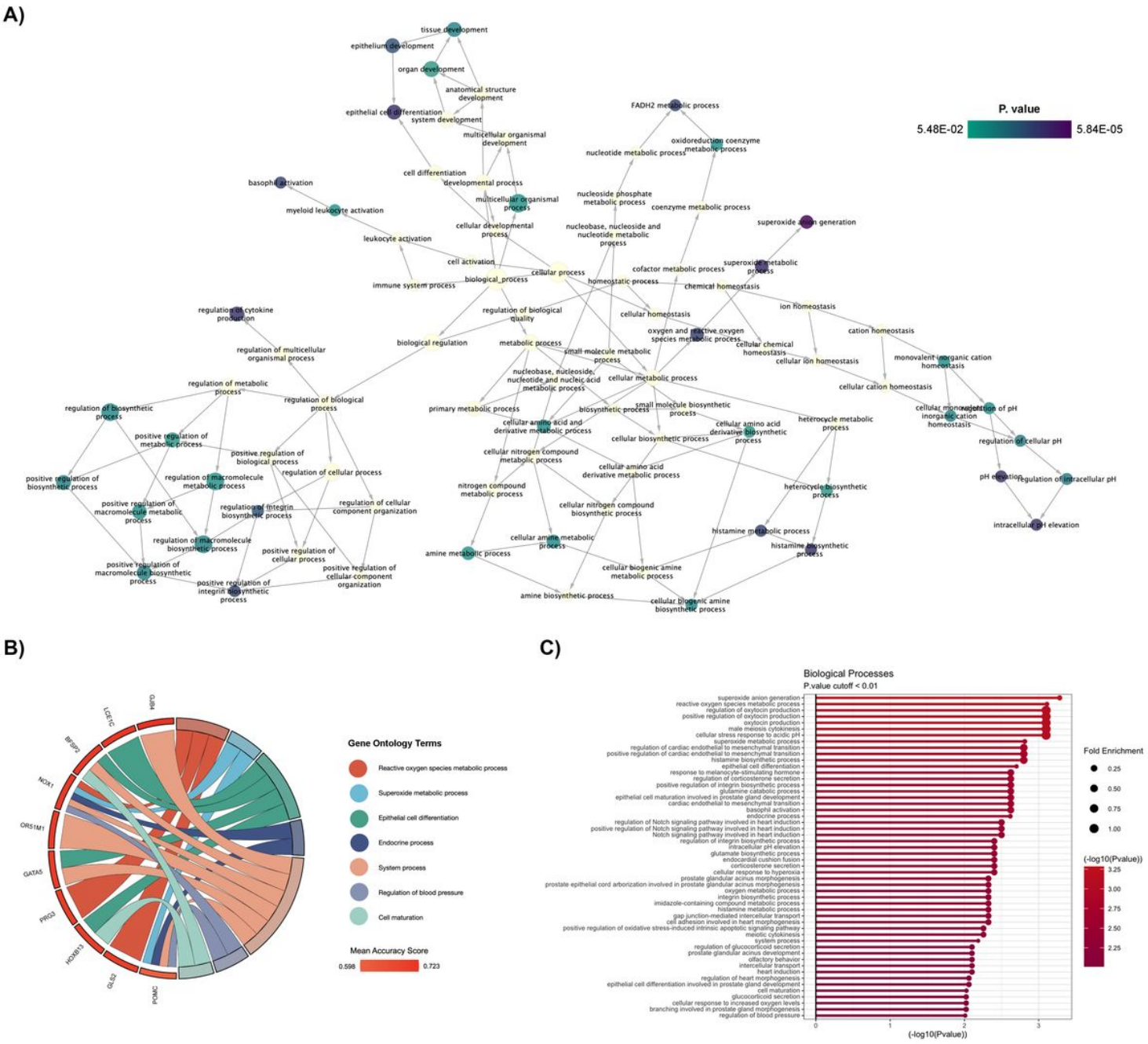


Figure 5

Biological Processes (BP) for KGs are relevant for oxidative stress and epithelial differentiation

Using Cytoscape BiNGO **A** we found gene enrichment of GO terms related to oxidative stress, pH regulation, epithelial differentiation, integrin biosynthesis, and histamine regulation processes using a cutoff of p. value <0.05. The network shows non significant processes in light yellow and in a color scale the significant terms. **B** To validate these results we run GOstat (p <0.01) on R and obtain 305 significant terms and we depict the terms that contain at least two of our KGs. The full GOstat result is shown in **C** where GO terms have been classified according to their relationship to each other based on Supplementary Figure 3.

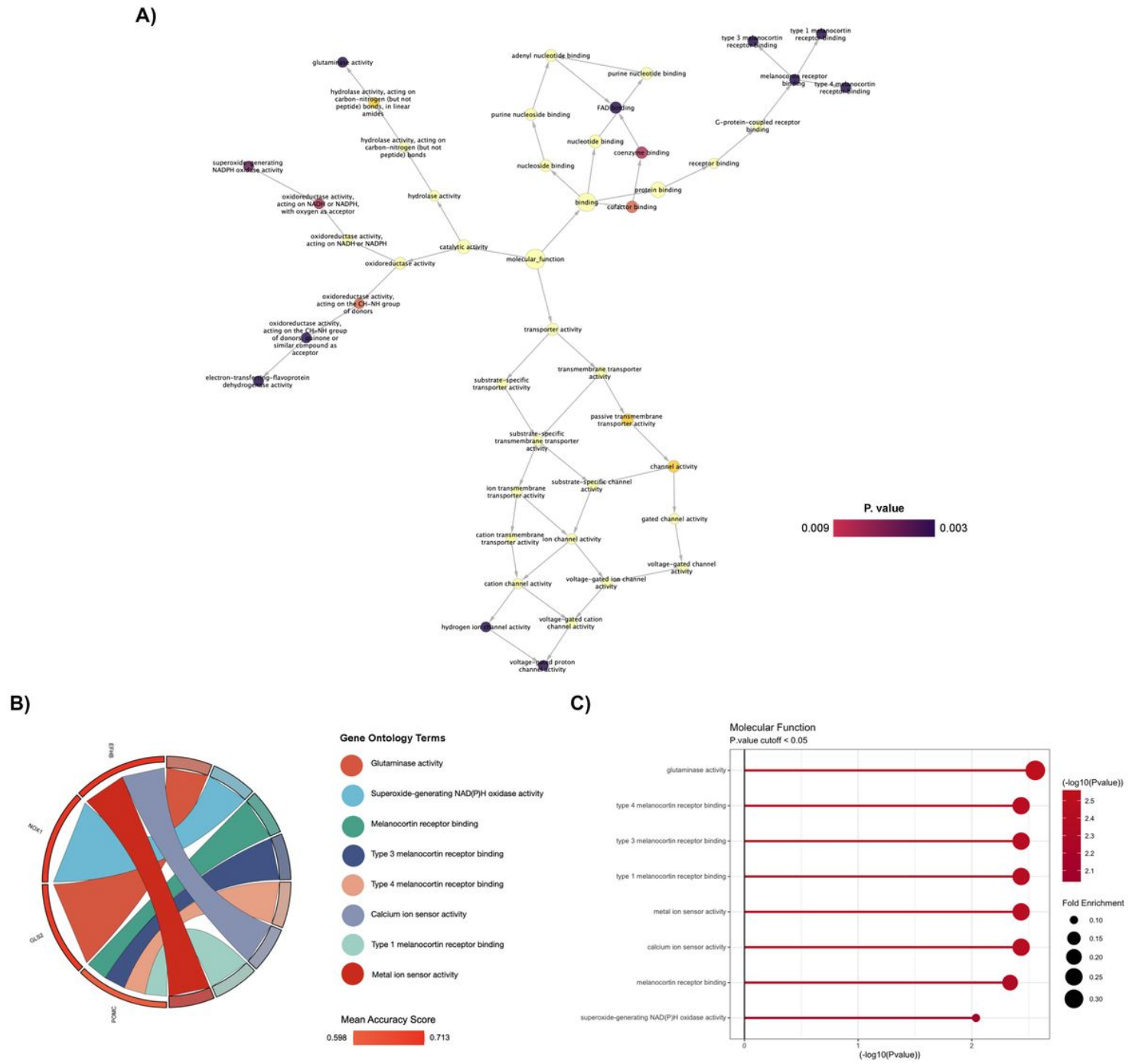


Figure 6

Molecular Functions (MF) using GOstat only for KG are relevant for NADPH activity, glutaminase activity and melanocortin receptor binding.

Using Cytoscape BiNGO **A** we found gene enrichment of GO terms related to NADPH activity, FAD binding, proton channels, melanocortin, and glutaminase activity using a cutoff of p. value <0.05. The network shows non-significant processes in light yellow and in a color scale the significant terms. **B** To validate these results we run GOstat (p <0.01) on R and obtain 8 significantly enriched terms. The full GOstat result is shown in **C** and relationship to each other based on Supplementary Figure 4.

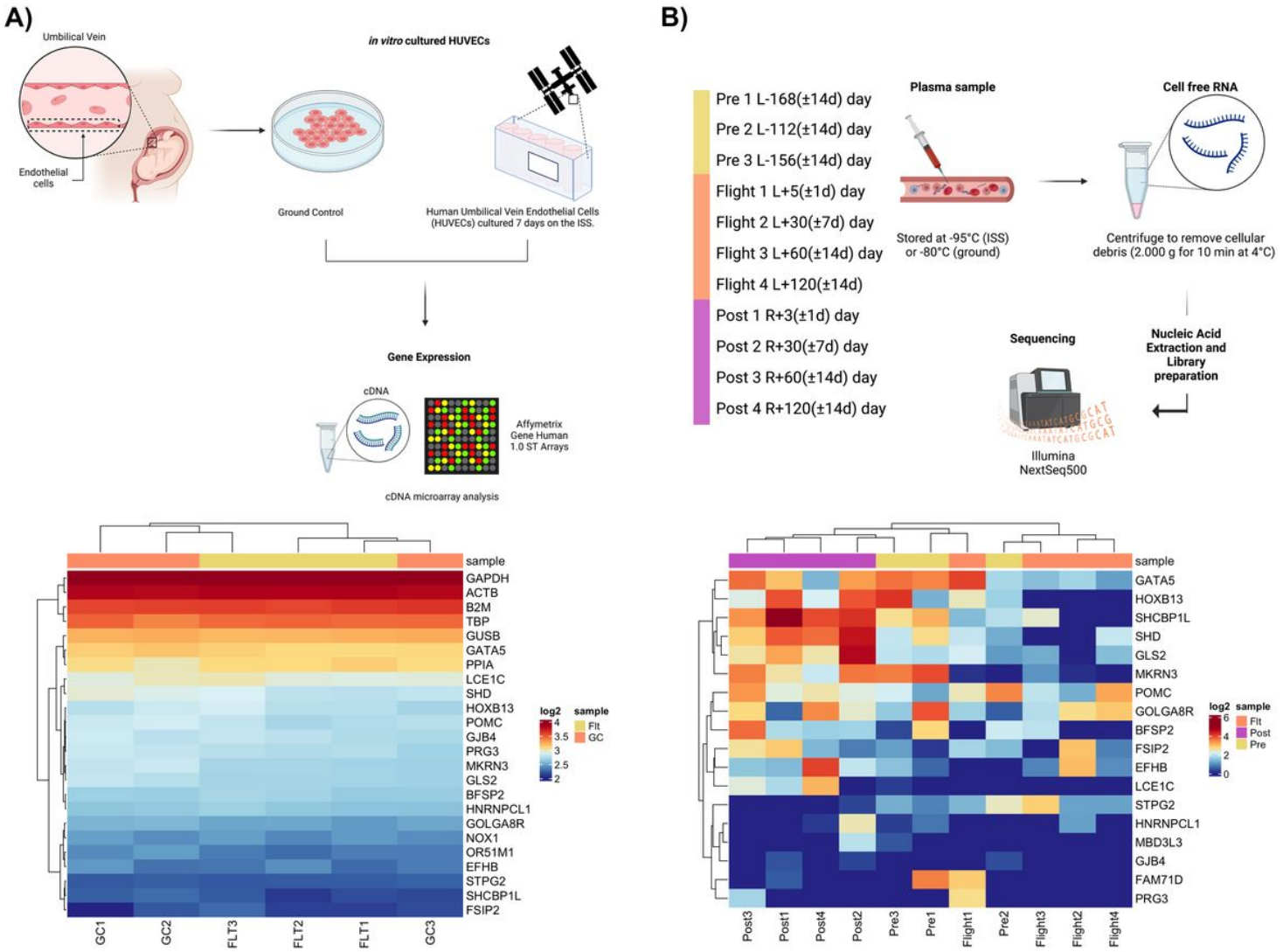


Figure 7

KGs are differentially expressed in both, sarcopenic patients and in the cell-free epigenome from JAXA, but not in the *in vitro* cultured HUVEC

A HUVEC OSD-52 FLT: Spaceflight, GC: Ground Control. **B** Gene expression from JAXA cell-free epigenome from OSD-530. This graph uses the same distance method and color information. For all graphs the log2 scale represents the log2 of the normalized counts for ease of comparison across genes.

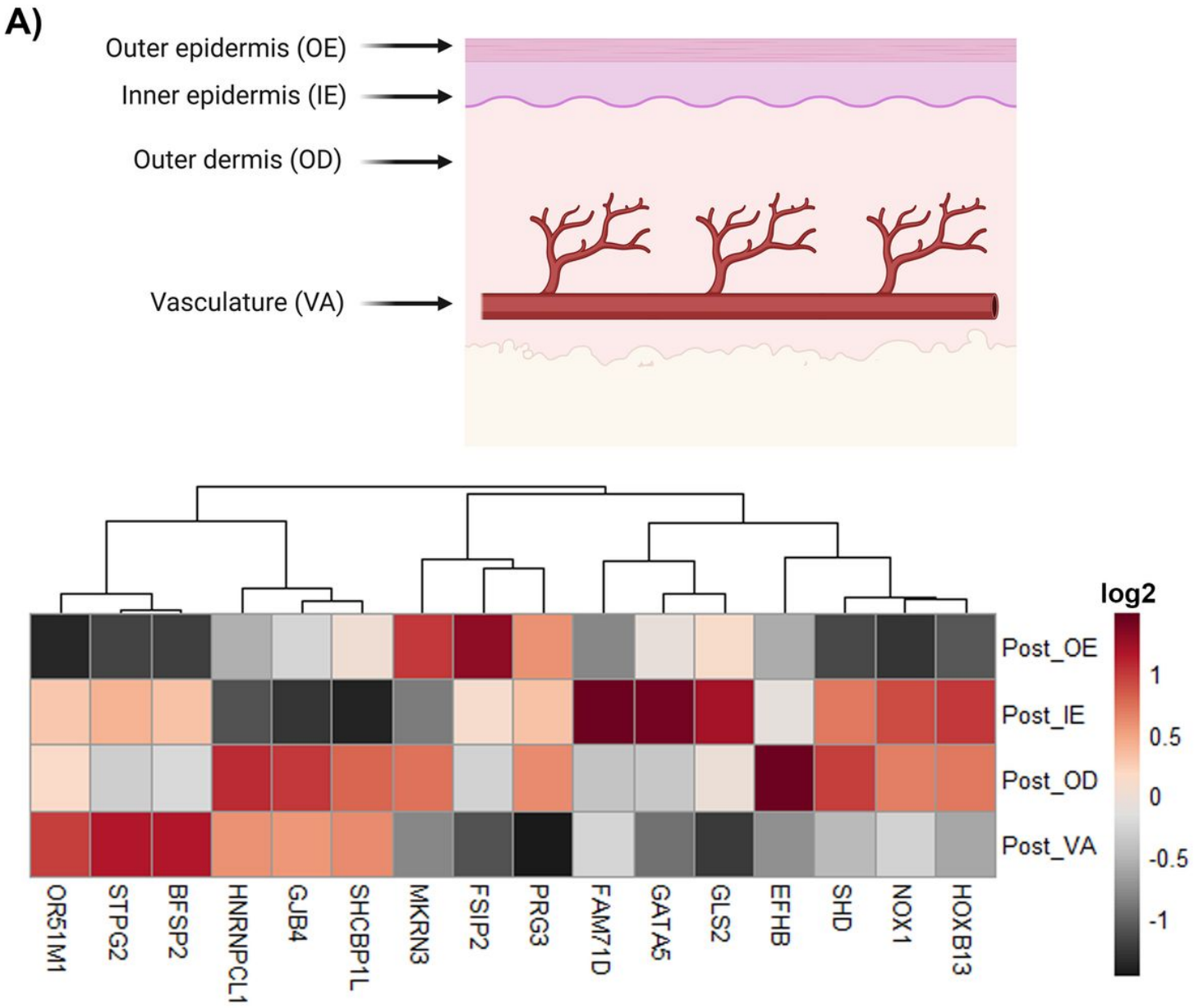


Figure 8

Response of sarcopenia markers in different skin compartments from I4 data

A Heatmap visualization of average scaled log fold changes of the gene sets associated with sarcopenia in the Inspiration4 postflight crew skin biopsy samples relative to preflight samples, by the skin compartment/region. OE = Outer epidermis, IE = Inner epidermis, OD = Outer Dermis, and VA = Vasculature.

Log FC

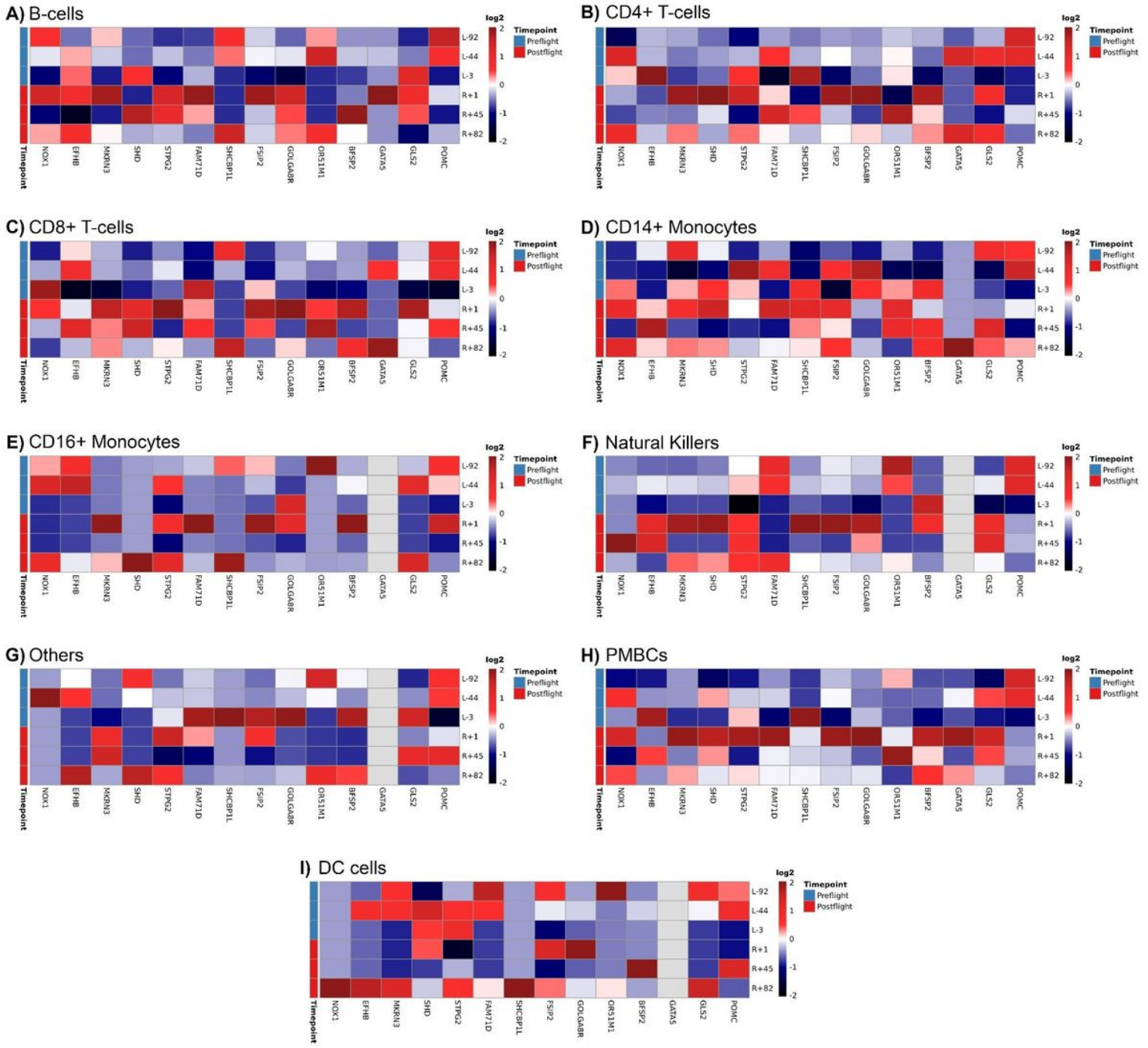


Figure 9

Response of sarcopenia markers in different blood cells from the i4 data

A B-cells **B** CD4+ T-cells **C** CD8+ T-cells **D** CD14+ Monocytes **E** CD16+ Monocytes **F** Natural Killers **G** Others: hematopoietic stem and pluripotent cell, red blood cell, Innate lymphoid cells, and platelets. **H** Pseudo-bulk Peripheral Blood Mononuclear Cells. **I** DC cells. The expression values were normalized. **Relative average expression - scaled by column**

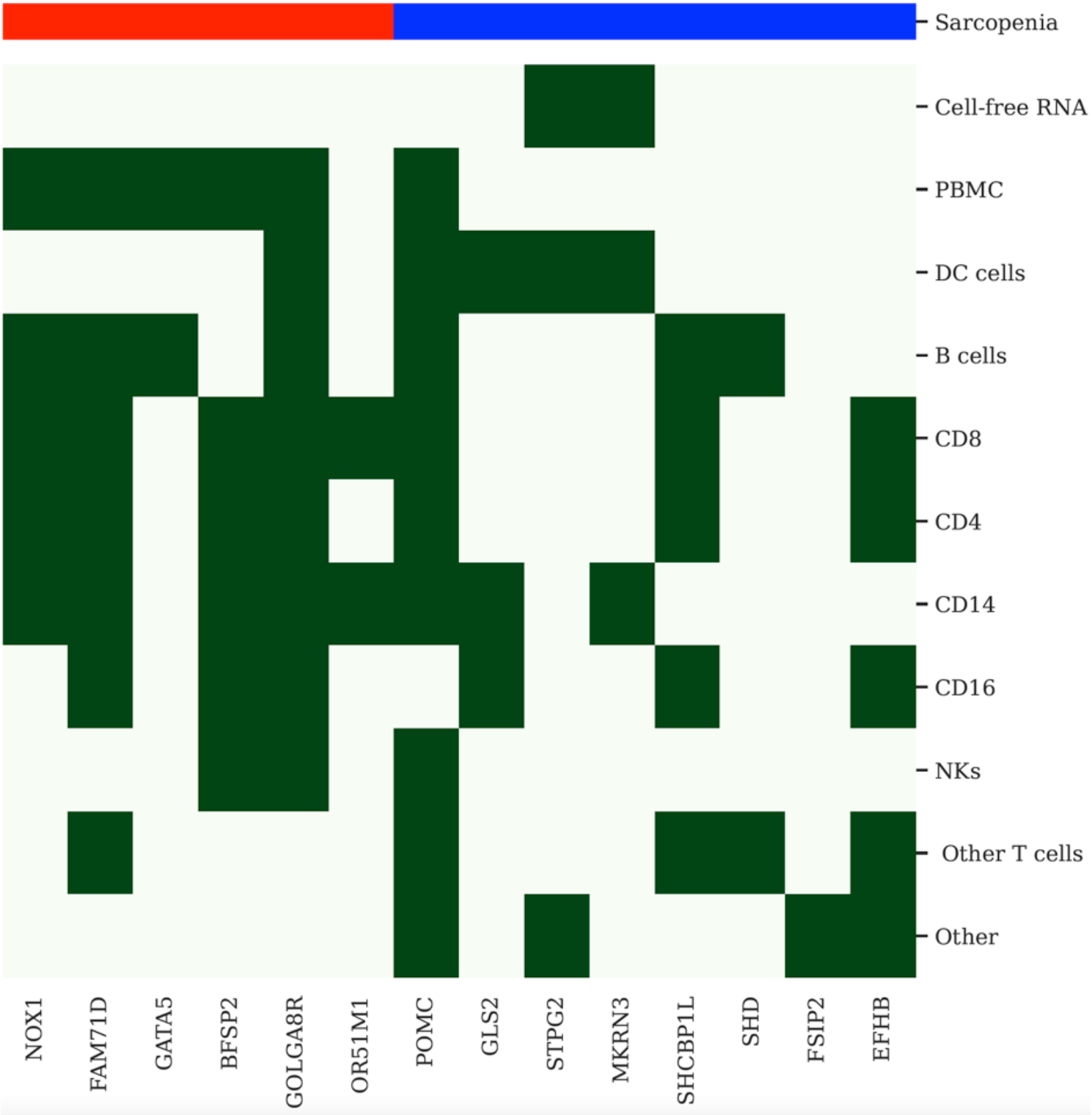


Figure 10

Genes present in all astronaut samples included in the analyzed databases.

The blue and red colors show whether genes from the sarcopenia database exhibited a decrease or increase in expression value, according to the corresponding fold change. This pattern of regulation is used to compare if these genes exhibit the same behavior across data from JAXA and I4. The green color indicates that a specific gene showed the same expression behavior as observed in the sarcopenia

dataset. Not all genes are displayed here due to excessive missing values in the expression data from JAXA or I4.

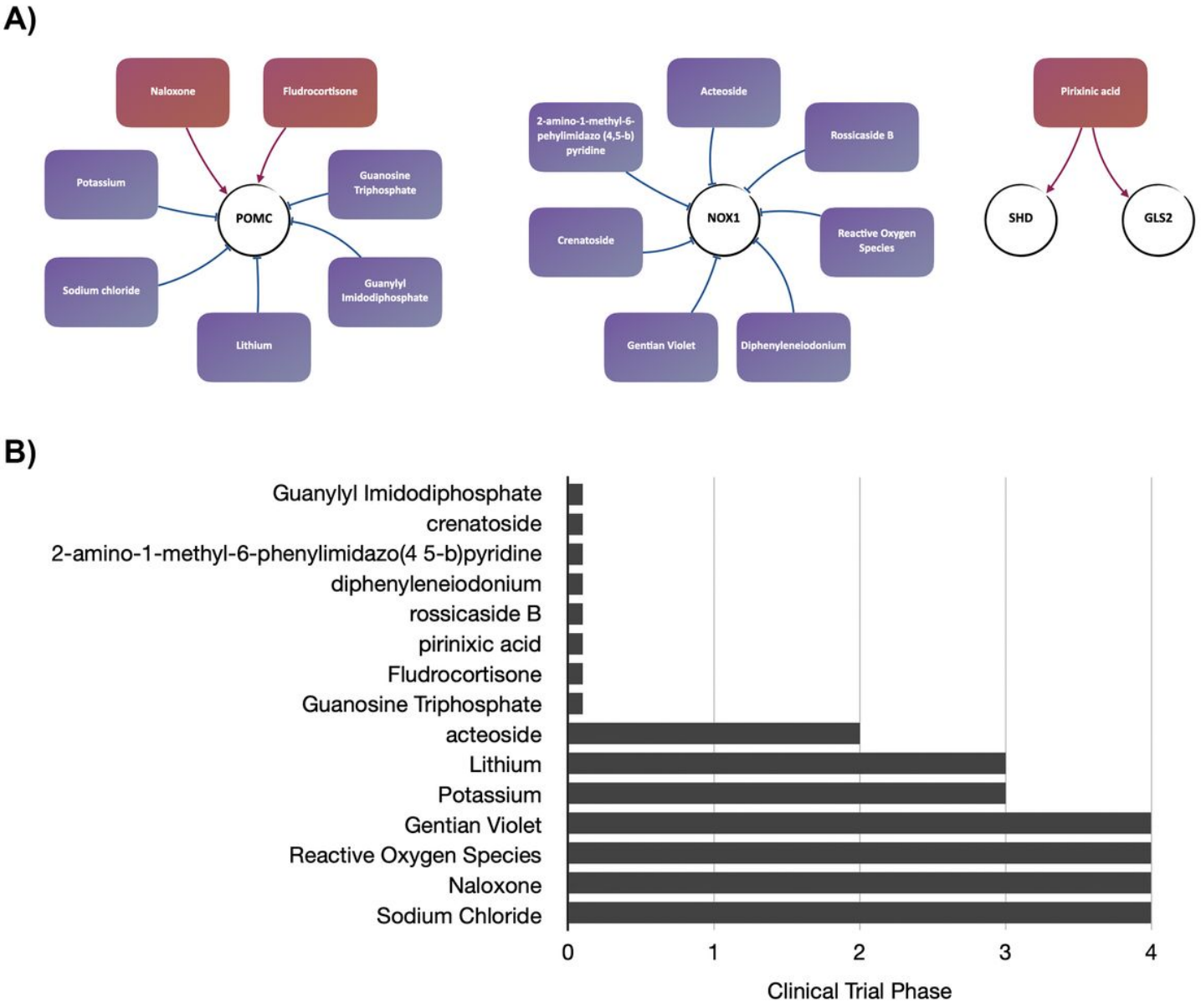


Figure 11

Chemicals interacting with four of the KGs from CTD

A Chemical Interactions were obtained from the CTD using only human evidence. We classified these chemicals based on their effect on the activity of our KGs. Red connecting lines represent the increased activity of the interacting protein, while blue means decreased activity of the interacting protein. KGs are represented in circles while drugs are shown in rectangles. **B** Clinical Trial Phase as reported in ChEMBL, all Phase IV have been approved.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- SupplementaryFigure1.png
- SupplementaryFigure2.png
- SupplementaryFigure3.png
- SupplementaryFigure4.png
- SupplementaryFigure5.png
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